MIL-C-46694A (MU)
28 April 1964
SUPERSEDING
MIL-C-46694 (MU)
25 April 1963

MILITARY SPECIFICATION

CHARGE, PROPELLING, 8 INCH, M1 (GREEN BAG) LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, 8 inch, M1, for 8 inch Howitzer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder Black
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC 1320

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DRAWINGS

ORDNANCE CORPS

8860491 - Charge, Propelling, 8 Inch: M1, For 8 Inch Howitzer, M2
76-5-85 - Container, Metal, Propelling Charge, Assembly
7548187 - Packing and Marking of Metal Container for Propelling Charges

PUBLICATIONS

ORDNANCE CORPS

IEL-6119358 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge.-The propelling charge shall comply with all requirements specified on Drawing (dwg.) 8860491 and with all requirements specified in applicable specification.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading shall not exceed the amount specified in MIL-P-223, when tested as specified in 4.3.1.

3.4 Propelling charge increment weight.-The charge increment weights shall comply with the weights specified in the loading authorization within the following tolerance.

Increment number	Tolerance (ounce)
1	.10
2	.05
3	.05
4	.05
5	.10

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3.5 Chamber gage.-The completed propelling charge shall freely enter the prescribed chamber gage.

3.6 Workmanship.-All components shall be free of dirt, grease, and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.6. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.
b. Number and revision date of specification.
c. Characteristic code number.
d. Lot or serial number.
e. Number of defectives found.
f. Assignable cause or analysis of occurrence.
g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of M1 Propelling Charges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same

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specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235, when applicable). Each lot shall contain:

a. Propellant from not more than one lot from one manufacturer.

b. Black powder of one interfix lot number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.40 percent for each minor defect and an AQL of 0.25 percent for each major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Charge 1 (see dwg. 8860504 covering a detail of dwg. 8860491)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Igniter charge assembly missing	Visual	01001 *
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that black powder or propellant can escape	Visual	01002
Minor:	AQL 1.50 percent		
201.	Any seam or opening improperly stitched	Visual	01003
202.	Thread broken	Visual	01004

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203.	Straps missing or improperly located	Visual	01005
204.	Evidence of poor workmanship (see 3.6)	Visual	01006

4.2.2.2 Charge 2, 3, 4 and 5 (see dwg. 8860502 covering a detail of dwg. 8860491)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	02001
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	02002
202.	Thread broken	Visual	02003
203.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.2.2.3 Charge assembly (see dwg. 8860491)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Total length maximum	Gage	03001
102.	Chamber gage failure	Gage	03002
103.	Assembly damaged to the extent that black powder or propellant can escape	Visual	03003
104.	Color of cloth improper	Visual	03004
Minor:	AQL 1.50 percent		
201.	Strap loose, broken or insecurely tied	Visual-Manual	03005
202.	Marking misleading or unidentifiable	Visual	03006
203.	Evidence of poor workmanship (see 3.6)	Visual	03007

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4.2.2.4 Container unsealed (see dwg. 76-5-85 and 7548187)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Any packing component missing or improperly assembled	Visual	04001
102.	Primer missing	Visual	04002
Minor:	AQL 1.00 percent		
201.	Charge cannot be removed by hand	Manual	04003
202.	Cover cannot be removed by hand	Manual	04004

4.2.2.5 Container sealed (see dwg. 76-5-85 and 7548187)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201.	Test plug missing or not greased	Visual	05001
202.	Test strip missing or torn	Visual	05002
203.	Car seal missing or incorrectly applied	Visual	05003
204.	Cover improperly engaged	Visual	05004
205.	Marking misleading or unidentifiable	Visual	05005

4.2.3 Testing

4.2.3.1 Moisture (see 3.3) Major defect - Code No. 06001.- The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of black powder from each eight hours production of

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of igniters. If the sample fails to comply with the applicable requirement the eight hours production represented by the sample shall be rejected.

4.2.3.2 Propelling charge (increment weight) (see 3.4) Major defect.-The charges shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

Propellant charge	Code No.
Increment one	07001
Increment two	07002
Increment three	07003
Increment four	07004
Increment five	07005

4.2.3.3 Air pressure test (see dwg. 7548187) Major defect.-The containers shall be checked 100 percent as specified in 4.3.2. If any container fails to meet the requirements it shall be rejected and removed from the lot.

4.2.4 Inspection equipment.-Index of Inspection Equipment Number IEL-6119358 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with MIL-P-223.

4.3.2 Air pressure test for container.-The container shall be subjected to a internal pressure of 3 pounds per square inch using test equipment in accordance with 4.2.4. A pressure drop equivalent to 3 inches of water in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The propelling charge shall be preserved in accordance with dwg. 7548187.

5.2 Packing

5.2.1 Level A.-The propelling charge shall be packed in a container complying with dwg. 76-5-85. Containers shall be closed, sealed, tested and marked in accordance with the requirements of dwg. 7548187.

5.3 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

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6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

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6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information

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necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Critical defect report

6.4.1 Submission of critical defect reports.-For U. S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP-RO; Picatinny Arsenal, ATTN: SMUPA-ND.

6.4.2 Reporting to other services.-For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

Custodian:
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Preparing activity:
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